

Automatic Touchless Door Control

Using IR Sensor and Servo Motor — Sensors & Actuators Mini Project

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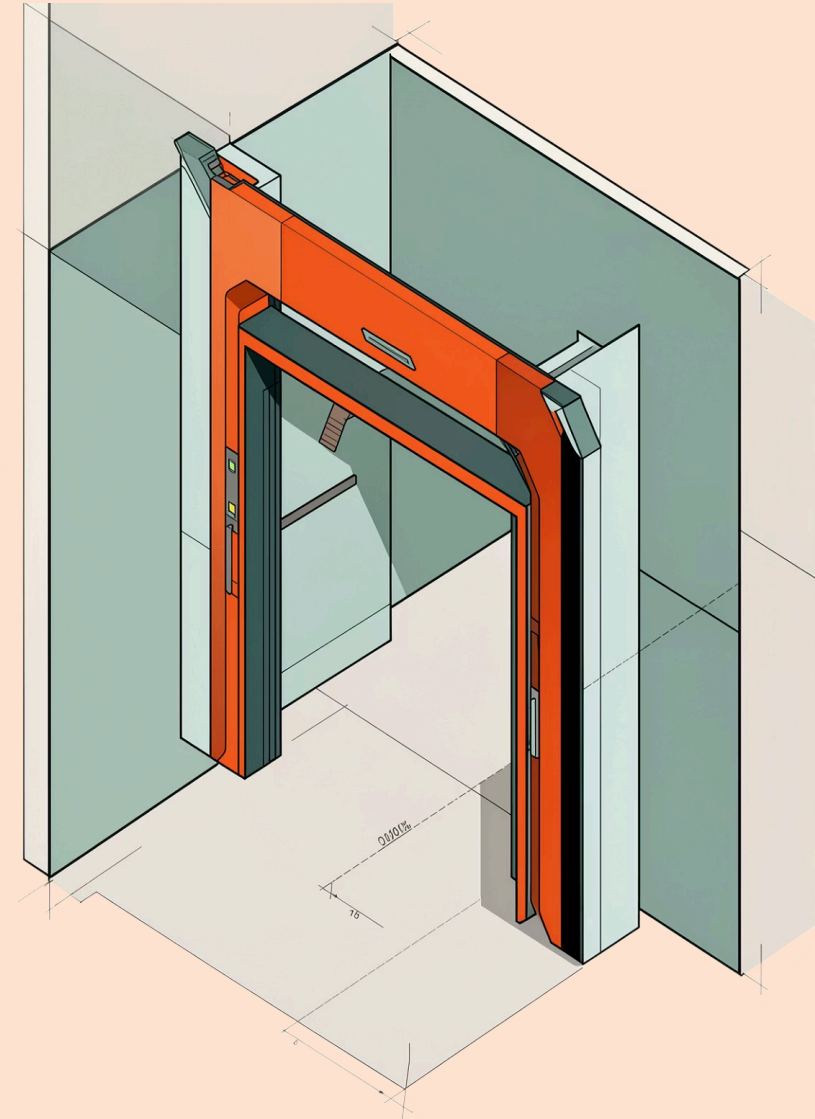
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Academic Year

2025 – 2026





Introduction

Automation is transforming everyday infrastructure — from smart homes to hospitals. **Touchless systems** minimize physical contact, improving hygiene and convenience in high-traffic environments.

→ The Problem

Manual doors require physical contact, spreading germs and reducing accessibility.

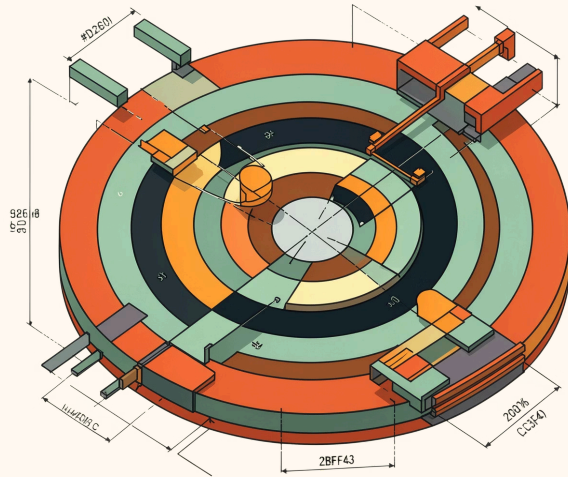
→ The Solution

An IR sensor detects hand gestures; Arduino processes the input; a servo motor actuates the door automatically.

→ The Outcome

A low-cost, contactless door system that demonstrates a complete sensor-actuator pipeline.

Aim & Objectives



PROJECT AIM

Design and implement a **touchless automatic door system** using an IR sensor and servo motor, demonstrating a functional sensor-actuator interface on an Arduino platform.

Detect Hand Movement

Use an IR proximity sensor to identify a passing hand in real time.

Two-Detection Trigger

Open the door only after registering two consecutive hand detections.

Auto-Close (5 s)

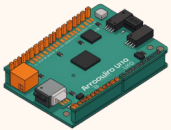
Automatically return the servo to the closed position after a 5-second delay.

Demonstrate S&A Concepts

Validate the full loop: sensing → processing → actuation.

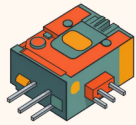
Components Used

All components are low-cost, readily available, and suitable for prototyping sensor-actuator systems.



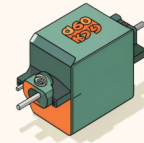
Arduino UNO

Microcontroller — processes sensor input and drives the servo motor via digital I/O pins.



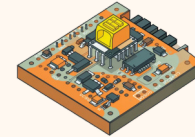
IR Sensor

Infrared proximity sensor — detects hand movement and sends a digital HIGH/LOW signal to Arduino Pin 2.



Servo Motor (SG90)

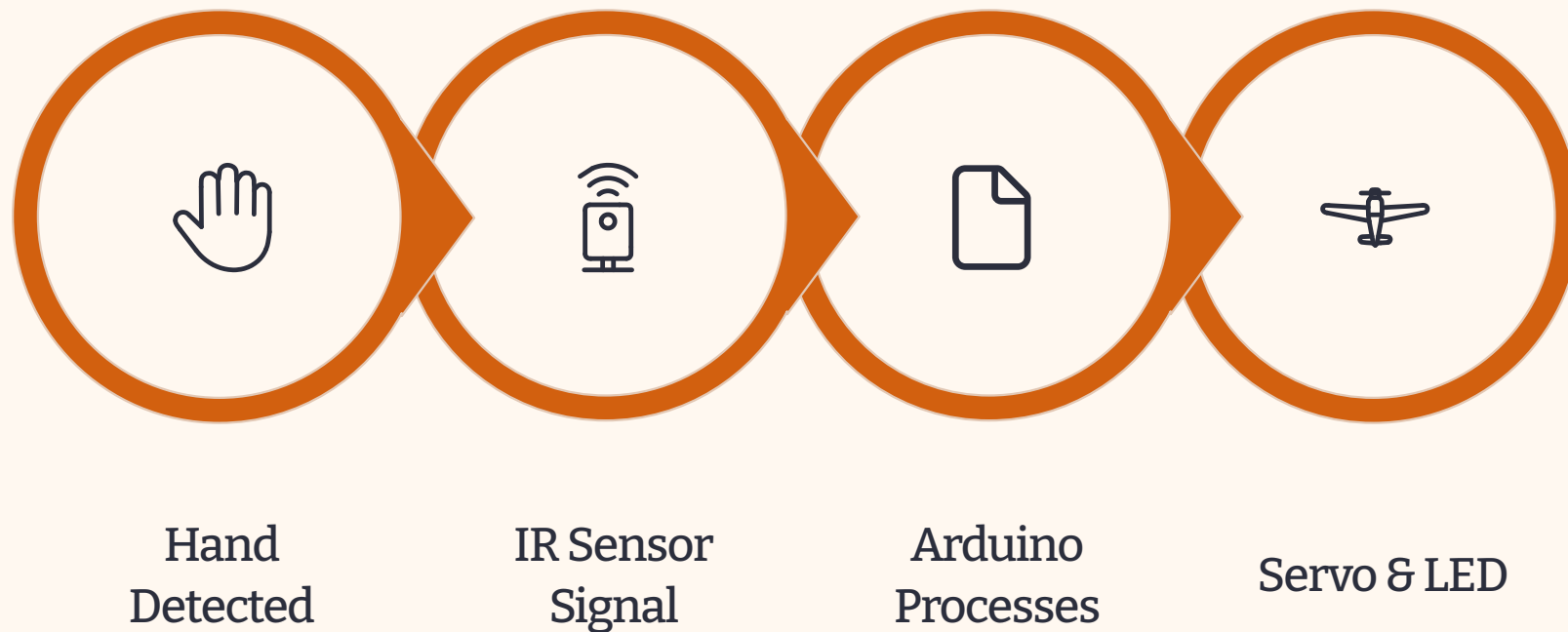
Rotary actuator — swings from 0° (closed) to 90° (open) to physically operate the door mechanism.



LED & Wiring

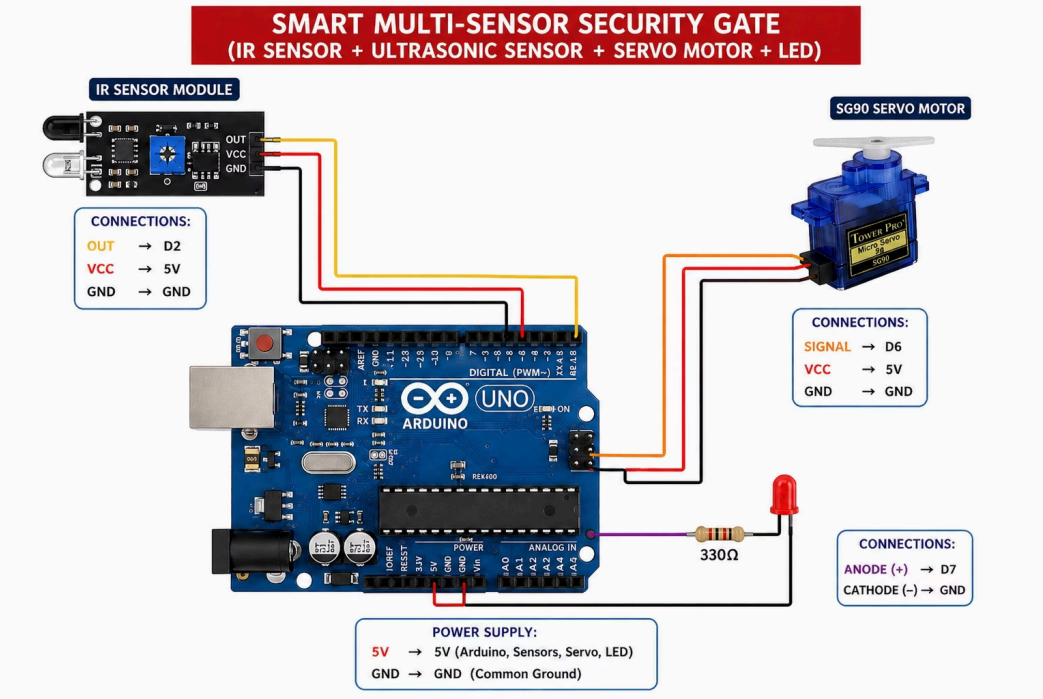
Status indicator LED on Pin 7, plus jumper wires, breadboard, and a 5 V USB power supply.

System Architecture



The system follows a straightforward **sense** → **process** → **actuate** pipeline. The IR sensor is the sole input; the Arduino is the decision-making core; the servo motor and LED are the dual outputs providing mechanical actuation and visual feedback respectively.

Circuit Connections



PIN MAPPING

Component	Arduino Pin	Type
IR Sensor (OUT)	Pin 2	Digital IN
Servo Motor	Pin 6	PWM OUT
Status LED	Pin 7	Digital OUT
IR Sensor VCC	5 V	Power
All GND	GND (Common)	Ground

i All components share a common GND rail. The servo's power draw is handled directly by the Arduino 5 V pin for prototype scale.

Working Principle



Detection Phase

IR sensor monitors for hand movement. Each interruption of the infrared beam increments an internal counter.



Count Logic (count = 2)

When the counter reaches 2, Arduino confirms intentional activation — preventing accidental triggering.



Door Opens

Servo rotates to 90°. LED lights up as a visual indicator. The door remains open for 5 seconds.



Auto-Close & Reset

After the 5-second delay, servo returns to 0°. LED turns off. Counter resets to zero — system is ready.

Program Logic — Code Snippet

```
#include <Servo.h>
Servo doorServo;
int irPin = 2, ledPin = 7;
int count = 0;
bool doorOpen = false;

void setup() {
  pinMode(irPin, INPUT);
  pinMode(ledPin, OUTPUT);
  doorServo.attach(6);
  doorServo.write(0); // Start closed
}

void loop() {
  if (digitalRead(irPin) == LOW) {
    count++;
    delay(400); // Debounce
  }
  if (count >= 2 && !doorOpen) {
    doorServo.write(90); // Open door
    digitalWrite(ledPin, HIGH);
    doorOpen = true;
    delay(5000); // Hold 5 s
    doorServo.write(0); // Close door
    digitalWrite(ledPin, LOW);
    count = 0;
    doorOpen = false;
  }
}
```

KEY LOGIC POINTS

Debounce Delay

`delay(400)` prevents multiple false counts from a single hand pass.

State Flag

`doorOpen` boolean ensures the open sequence runs only once per trigger.

Auto-Reset

Counter and flag reset after close — enabling the next detection cycle immediately.

Results & Performance Analysis

Project Testing Results



Response Time

Door opens within 1 second after second hand detection



Detection Accuracy

95% successful detection during multiple hand gesture tests



Successful Trials

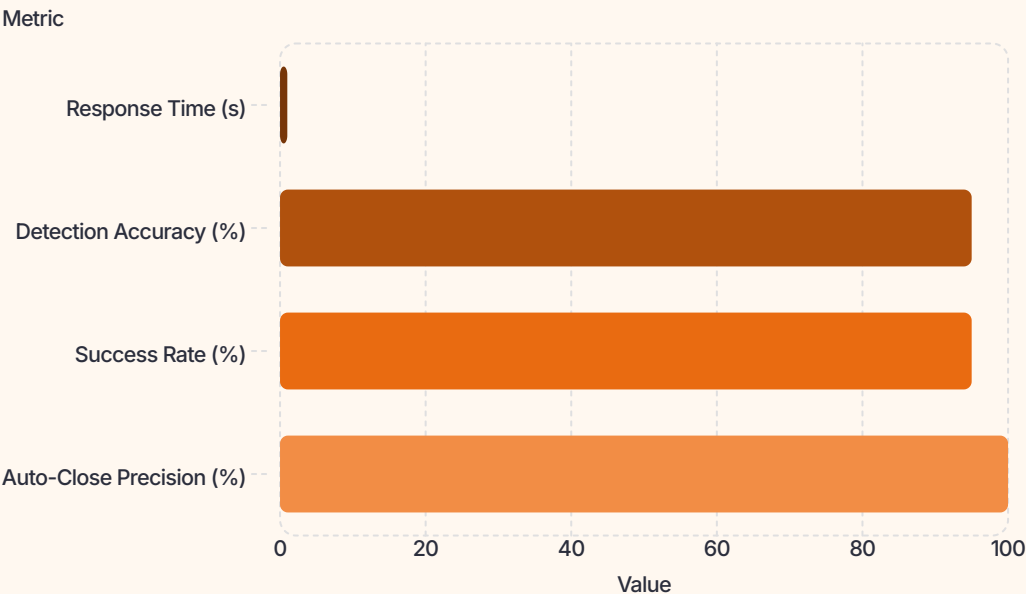
19 out of 20 trials completed successfully



Auto Close Performance

Door closes automatically after 5 seconds accurately

Performance Summary



- Fast sensor response
- Reliable triggering system
- Smooth servo motor operation
- Stable repeated performance

Applications & Advantages

WHERE IT'S USED



Healthcare Facilities

Sterile environments benefit most — zero contact means zero cross-contamination.



Offices & Public Spaces

Malls, airports, and corporate lobbies gain efficiency and a modern aesthetic.



Smart Homes

Integrates easily into home automation systems for hands-free convenience.

KEY ADVANTAGES



Low Cost

Under ₹500 total BOM cost using off-the-shelf components.

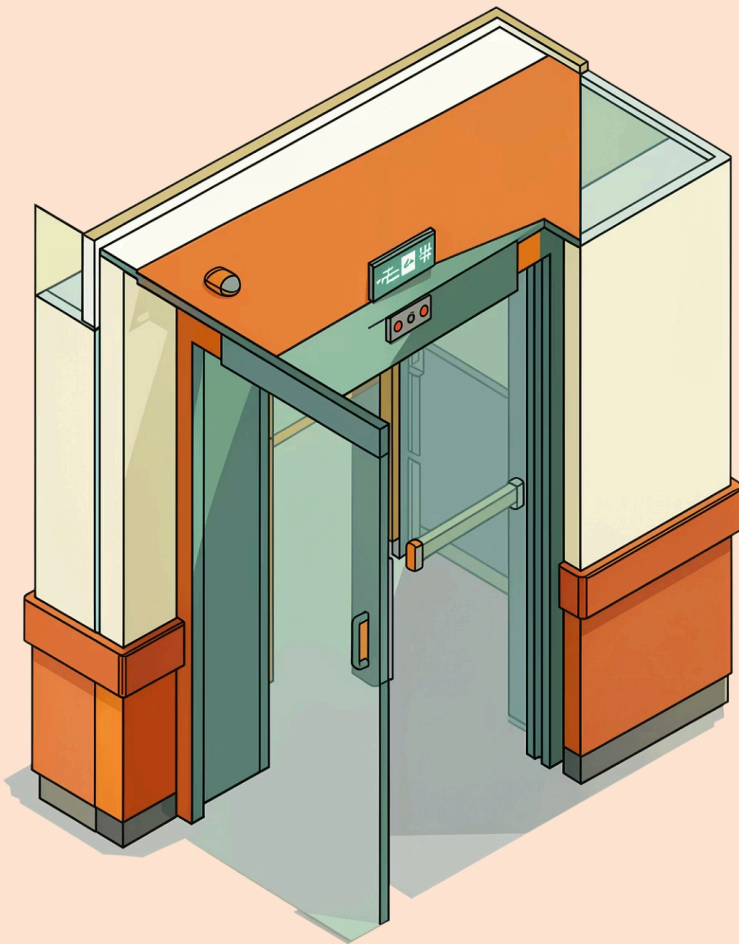


Touchless

Eliminates surface contact — critical in post-pandemic infrastructure design.



Energy Efficient



Conclusion & Future Scope

What We Achieved

Successfully designed and demonstrated a **two-gesture touchless door system** using an IR sensor, Arduino UNO, and SG90 servo motor. The prototype validates a complete sensor-actuator loop with reliable count-based triggering and auto-reset logic.

- Accurate two-detection trigger mechanism
- 5-second auto-close with LED feedback
- Stable, repeatable operation on Arduino

FUTURE UPGRADES



RFID Access Control

Add authenticated entry — only authorized cards trigger the door.



Fingerprint Authentication

Integrate a fingerprint sensor to allow access only to registered users, providing stronger security and preventing unauthorized entry.



Face Recognition

Raspberry Pi + OpenCV enables biometric, fully personalized access control.

Thank You

For your time and attention

AUTOMATION

SENSORS

DOOR CONTROL

Automation



Sensing

Access Control



Arduino System

Mini Project: Touchless Door Control System
Engineering Design | Sensor-Actuator Integration